

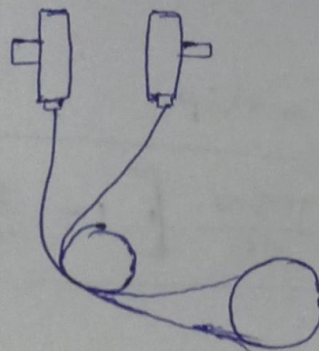
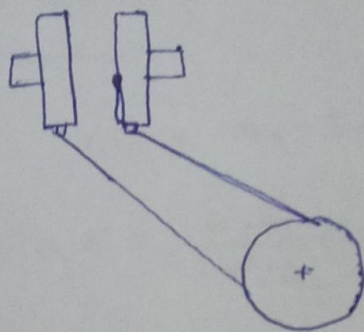
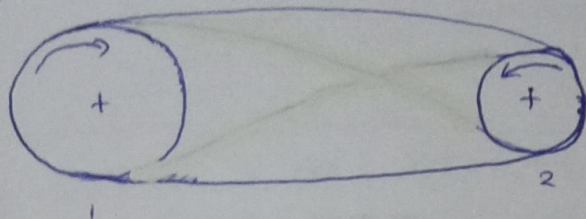
Belt, rope, chain.

Driver

Driven.

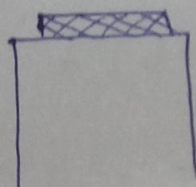
belt.

← Open system.

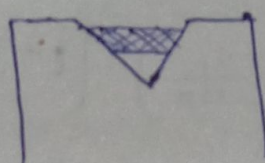


Quarter turn
belt system.

Flat belt



V-belt.

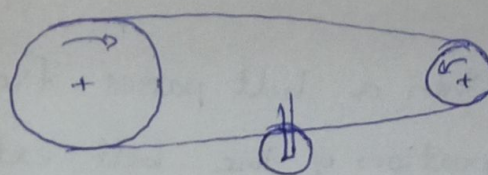


Circular/Rotor belt.



Other types of belts are:

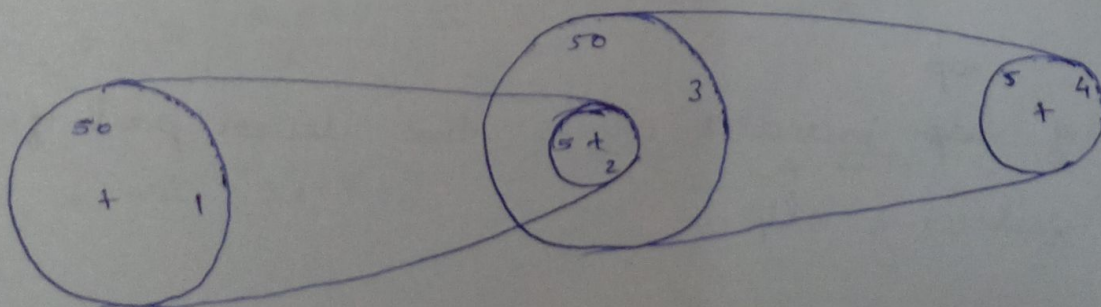
- Leather belt.
- Cotton, belt fabric
- Rubber belt.
- Belata belt.



When thick belt is used, we consider the thickness of belt till its centre.

$$\frac{N_2}{N_1} = \frac{D_1}{D} \Rightarrow \frac{N_2}{N_1} = \frac{D_1 + t/2 + t/2}{D_2 + t/2 + t/2}$$

ie, $\boxed{\frac{N_2}{N_1} = \frac{D_1 + t}{D_2 + t}}$



$$N_2 = \frac{50}{5} \times 100 = \underline{\underline{1000}} \text{ rpm.}$$

$$\frac{N_4}{N_1} = \frac{D_1 \times D_5}{D_2 \times D_4}$$

$$\text{Velocity, } v = \frac{\pi D N}{60}$$

$$v_b \text{ at driven} = \frac{\pi D N}{60} - \frac{s_1}{100} \times \frac{\pi D N}{60}$$

where, $s \Rightarrow$ slip.

$$v_b = \frac{\pi D N}{60} \left[1 - \frac{s_1}{100} \right]$$

$$\frac{\pi D_2 N_2}{60} = v - \frac{s_2}{100} v = v \left(1 - \frac{s_2}{100} \right)$$

$$= \frac{\pi D_1 N_1}{60} \left[1 - \frac{s_1}{100} \right] \left[1 - \frac{s_2}{100} \right]$$

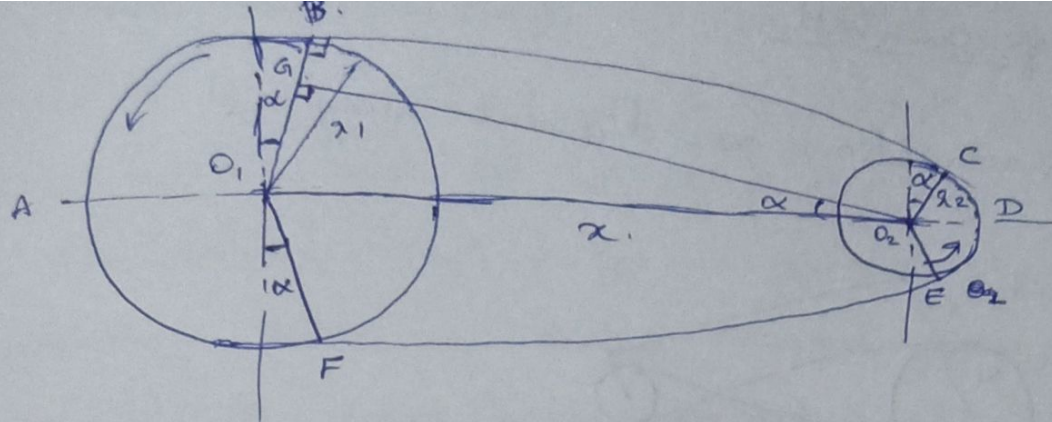
$$N_2 = N_1 \frac{D_1}{D_2} \left[1 - \frac{s_1}{100} \right] \left[1 - \frac{s_2}{100} \right]$$

When a belt passes from slack side to tight side, a certain portion of the belt extends & it contracts again when the belt passes from the tight side to slack side. Due to these changes of length, there is a relative motion b/w the belt and pulley surface. This relative motion is termed as creep. The total effect of creep is to reduce slightly the speed of driven pulley.

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \frac{E + \sqrt{\sigma_1}}{E + \sqrt{\sigma_2}}$$

wrap.

→ Angle of ~~lap~~ increases when the lower part of belt is kept tight reducing slip.



$$L \text{ of belt} = 2 \text{ arc } AB + BC + \text{arc } CD.$$

$$L = 2\theta$$

$$\theta_1 = \frac{\pi}{2} + \alpha, \quad \theta_2 = \frac{\pi}{2} - \alpha.$$

$$OG = r_1 - r_2$$

$$\sin \alpha = \frac{OG}{O_1O_2} = \frac{r_1 - r_2}{x}.$$

$$BC = GO_2 = \sqrt{x^2 - (r_1 - r_2)^2}$$

$$= (x^2 - (r_1 - r_2)^2)^{1/2}$$

$$\therefore BC = x \left[1 - \frac{(r_1 - r_2)^2}{x^2} \right]^{1/2}.$$

Using binomial theorem,

$$BC = x \left[1 - \frac{1}{2} \left(\frac{r_1 - r_2}{x} \right)^2 + \dots \right]$$

$$BC = x - \frac{1}{2} \frac{(r_1 - r_2)^2}{x}$$

$$L = 2 \left[r_1 \frac{\pi}{2} + r_1 \alpha + x - \frac{(r_1 - r_2)^2}{2x} + r_2 \frac{\pi}{2} - r_2 \alpha \right]$$

$$= 2 \left[(r_1 + r_2) \frac{\pi}{2} + \alpha (r_1 - r_2) + x - \frac{(r_1 - r_2)^2}{2x} \right]$$

$$= 2 \left[\frac{(r_1 + r_2)}{2} \pi + \frac{(r_1 - r_2)}{x} (r_1 - r_2) + x - \frac{(r_1 - r_2)^2}{2x} \right]$$

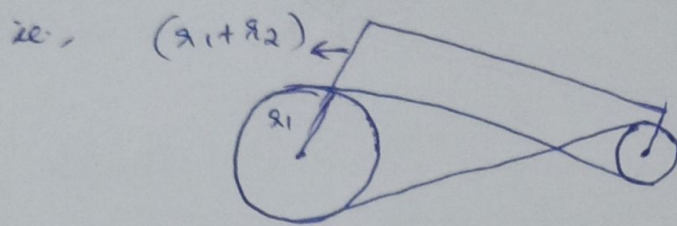
$$= 2 \left[\frac{(r_1 + r_2)}{2} \pi + \frac{(r_1 - r_2)^2}{x} - \frac{(r_1 - r_2)^2}{2x} + x \right]$$

$$= 2 \left[\frac{(r_1 + r_2)}{2} \pi + \frac{(r_1 - r_2)^2}{x} + x \right]$$

$$\therefore L = \pi (r_1 + r_2) + \frac{(r_1 - r_2)^2}{x} + 2x.$$

Length of cross belt.

Cross belt construction goes different way.

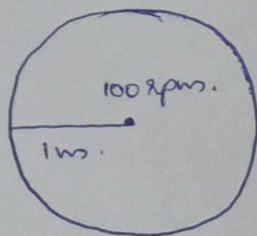


hence, while driving:

$$L = \pi (r_1 + r_2) + \frac{(r_1 + r_2)^2}{x} + 2x$$

7-2-25.

Slip of belt.



$$N = 100 \text{ rpm.}$$

$$r = 1 \text{ m.}$$

$$V = \frac{\pi D N}{60}$$

$$L = r\theta$$

$$\frac{dL}{dt} = r \frac{d\theta}{dt}$$

$$V = r\omega = r \frac{\pi N}{60}$$

$$\therefore V = \frac{\pi D N}{60}$$

ie

$$V = \frac{\pi D N}{60} = \frac{s}{100} \times \frac{\pi D N}{60}$$

$$V = \frac{\pi D N}{60} \left[1 - \frac{s_1}{100} \right]$$

$$\frac{\pi D_2 N_2}{60} = V - \frac{s_2}{100} V$$

$$\frac{\pi D_2 N_2}{60} = V \left[1 - \frac{s_2}{100} \right]$$

$$\frac{\pi D_2 N_2}{60} = \frac{\pi D_1 N_1}{60} \left[1 - \frac{s_1}{100} \right] \left[1 - \frac{s_2}{100} \right]$$

$$\therefore N_2 = \frac{D_1}{D_2} N_1 \left[1 - \frac{s_1}{100} \right] \left[1 - \frac{s_2}{100} \right]$$

1. A shaft which rotates at constant speed of 160 rpm is connected by a belting to a parallel shaft 720 mm apart which has to run at 60, 80 & 100 rpm. The smallest pulley on the driving shaft is 40 mm in radius. Determine the remaining radii of the 2 stepped pulleys for (i). cross belt. (ii). an open belt. Neglect belt thickness & slip.

Ans. $\theta = \pi - 2\alpha$ $\rightarrow$ Angle of ^{wrap} ~~lap~~ is small.

Cross belt $\rightarrow \theta = \pi + 2\alpha$

$$N_1 = 160 \text{ rpm.}$$

$$d_p = 80 \text{ mm.} \Rightarrow r_p = 40 \text{ mm}$$

$$V = \frac{\pi D N}{60} = \frac{\pi \times 80 \times 160}{60} = \underline{\underline{670.2 \text{ mm/s.}}}$$

$$= \underline{\underline{0.67 \text{ m/s}}}$$

We know that, $\frac{N_2}{N_1} = \frac{D_1}{D_2}$

$$D_1 + D_2 = D_3 + D_4 = D_5 + D_6$$

$$N_2 = 60 \text{ rpm, } N_3 = 80 \text{ rpm, } N_4 = 100 \text{ rpm.}$$

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \Rightarrow \frac{60}{160} = \frac{80}{D_2} \Rightarrow D_2 = 213.3 \text{ mm} = \underline{\underline{0.21 \text{ m}}}$$

$$\frac{r_2}{r_1} = \frac{N_1}{N_2} \Rightarrow r_2 = \frac{160}{60} \times 40 = 106.6 \text{ mm} = \underline{\underline{0.106 \text{ m}}}$$

$$\frac{N_3}{N_4} = \frac{160}{80} = \underline{\underline{2 \text{ rpm.}}}$$

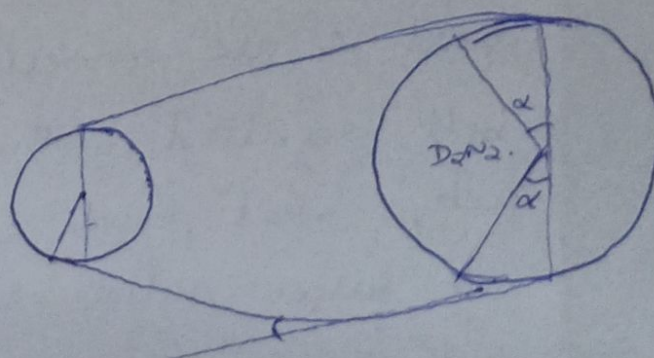
Power = rate of work done.
Tension from in the belt $\Rightarrow$ force.

Tight - T_1

Slack - T_2

$$\text{Net force} = T_1 - T_2$$

$$P = (T_1 - T_2) u.$$



$$\frac{T_1}{T_2} = e^{\mu\theta}$$

$T_1 \Rightarrow$ Tight side tension.

$T_2 \Rightarrow$ slack side tension.

$\theta \Rightarrow$ rad.

$$\theta = (180 - 2\alpha^\circ) \frac{\pi}{180}$$

$$\theta = (180 + 2\alpha^\circ) \frac{\pi}{180}$$

$\mu =$ coefficient of friction.

2. Find the power transmitted by a belt running over a pulley of 600mm dia at 200 rpm. $\mu = 0.25$ & of lap 160° . Max tension in the belt is 2500N

Ans. $P = (T_1 - T_2) \cdot v$

$$\text{lap} = 160 \times \frac{\pi}{180} = 2.79 \text{ rad}$$

$$T_1 = 2500 \text{ N}$$

$$\frac{T_1}{T_2} = e^{\mu\theta} \Rightarrow \frac{2500}{T_2} = e^{(0.25 \times 2.79)} = \underline{\underline{2.008}}$$

$$\therefore T_2 = \frac{2500}{2.008} = \underline{\underline{1250.47 \text{ N}}}$$

$$v = \frac{\pi DN}{60} = \frac{\pi \times 0.6 \times 200}{60} = \underline{\underline{6.28 \text{ m/s}}}$$

$$\therefore P = (2500 - 1250.47) \times 6.28 = \underline{\underline{7850 \text{ W}}}$$

3. 2 pulleys 450 & 200 mm dia are ^{on} parallel shaft 1.95m apart & are connected by cross belt. Find the length of the belt required & the angle of contact b/w belt & each pulley. What power can be transmitted by the belt when the larger pulley rotates at 200 rpm. The max permissible tension in the belt is 1kN & $\mu = 0.25$.

Ans. $D_1 = 450 \text{ mm}$

$T_1 = 1 \text{ kN} = 1000 \text{ N}$

$N_1 = 200 \text{ rpm}$

$D_2 = 200 \text{ mm}$

$\mu = 0.25$

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \Rightarrow \frac{N_2}{200} = \frac{450}{200} \Rightarrow N_2 = \frac{450}{200} \times 200$$

$$\therefore N_2 = \underline{\underline{89 \text{ rpm}}} \quad N_2 = 200$$

$$\begin{aligned} L &= \pi (r_1 + r_2) + \frac{(r_1 + r_2)^2}{x} + 2x \\ &= \pi (0.225 + 0.100) + \frac{(0.225 + 0.100)^2}{1.95} + (2 \times 1.95) \\ &= 1.02 + 0.054 + 3.9 \\ &= \underline{\underline{4.97 \text{ m}}} \end{aligned}$$

$$\text{Angle of contact} = (180 + 2\alpha) \frac{\pi}{180}$$

$$\begin{aligned} \sin \alpha &= \frac{r_1 - r_2}{x} \quad \text{for open belt.} \\ \sin \alpha &= \frac{r_1 + r_2}{x} \quad \text{for cross belt.} \end{aligned}$$

$$\sin \alpha = \frac{0.225 + 0.1}{1.95} = 0.166$$

$$\therefore \alpha = \sin^{-1} 0.166 = \underline{\underline{9.59^\circ}}$$

$$\text{Angle of contact} = (180 + 9.59 \times 2) \frac{\pi}{180} = \underline{\underline{3.47}}$$